

StockMaster — Software Requirements Specification (SRS)

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Project: StockMaster — Professional Inventory Management System

1. Introduction

1.1 Purpose

This document describes the functional and non-functional requirements for StockMaster v4.0, a desktop inventory management application designed for small to medium businesses.

1.2 Scope

StockMaster provides stock control, point of sale (POS), purchase order management, reporting, user authentication, and bilingual support (Spanish/English) — all running locally without internet dependency.

1.3 Definitions

Term	Definition
POS	Point of Sale — module for processing customer sales
IVA	Impuesto al Valor Agregado — Argentine value-added tax
Batch	A specific lot of a product with its own expiry date and quantity
KPI	Key Performance Indicator — metrics shown on the dashboard
i18n	Internationalization — multi-language support
ESC/POS	Epson Standard Code for Point of Sale — thermal printer protocol

2. Overall Description

2.1 Product Perspective

StockMaster is a standalone desktop application built with JavaFX. It uses an embedded SQLite database for local data persistence, requiring no external server or internet connection.

2.2 User Classes

Role	Permissions
ADMIN	Full access: manage products, users, suppliers, categories, sales, purchase orders, reports, backup/restore, language settings
CASHIER	Limited access: sales (POS), product search, view dashboard

2.3 Operating Environment

- OS:** Windows 10/11 (64-bit)
- Runtime:** Java 17+ (bundled via jpackage installer)
- Database:** SQLite (embedded, file-based)
- Hardware:** Optional ESC/POS thermal printer for receipts

2.4 Design Constraints

- Must work fully offline
- All data stored locally (privacy by design)
- Single-user at a time (desktop app)

- Installer under 100 MB

3. Functional Requirements

3.1 Authentication (FR-AUTH)

| ID | Requirement | Priority |

|---|-----|-----|

| FR-AUTH-01 | System shall display a login screen on startup | High |

| FR-AUTH-02 | Users authenticate with username and password (SHA-256 hashed) | High |

| FR-AUTH-03 | Default admin account created on first run (admin/admin) | High |

| FR-AUTH-04 | System tracks current session (username, role, user ID) | High |

| FR-AUTH-05 | Logout returns to login screen (Ctrl+Q shortcut) | Medium |

3.2 Dashboard (FR-DASH)

| ID | Requirement | Priority |

|---|-----|-----|

| FR-DASH-01 | Display KPI cards: total products, inventory value, low stock count, expiring batches (30 days) | High |

| FR-DASH-02 | Display bar chart of product distribution by category | Medium |

| FR-DASH-03 | KPI values update on view load | High |

3.3 Product Management (FR-PROD)

| ID | Requirement | Priority |

|---|-----|-----|

| FR-PROD-01 | CRUD operations for products (name, description, category, supplier, prices, stock, barcode, tax rate) | High |

| FR-PROD-02 | Search/filter products by name, category, or barcode | High |

| FR-PROD-03 | Display products in sortable table with columns: ID, Name, Category, Supplier, Buy Price, Sell Price, Stock, Min Stock | High |

| FR-PROD-04 | Visual low-stock indicators (row highlighting) | Medium |

| FR-PROD-05 | Barcode detection via keyboard input | Medium |

| FR-PROD-06 | Import products from CSV file | Medium |

| FR-PROD-07 | Export products to CSV file | Medium |

| FR-PROD-08 | Export product list to PDF | Medium |

| FR-PROD-09 | Print product ticket to ESC/POS thermal printer | Low |

| FR-PROD-10 | Open cash drawer via ESC/POS command | Low |

3.4 Batch/Lot Tracking (FR-BATCH)

| ID | Requirement | Priority |

|---|-----|-----|

| FR-BATCH-01 | Each product can have multiple batches with: lot number, expiry date, quantity, cost | High |

| FR-BATCH-02 | Batch panel appears when selecting a product in the table | High |

| FR-BATCH-03 | Add, delete batches inline | High |

| FR-BATCH-04 | Display batch status (ACTIVE, EXPIRED) based on expiry date | Medium |

3.5 Sales / Point of Sale (FR-SALE)

| ID | Requirement | Priority |

|---|-----|-----|

| FR-SALE-01 | Search products by name or barcode for adding to cart | High |

| FR-SALE-02 | Shopping cart with product name, quantity, unit price, tax rate, line total | High |

| FR-SALE-03 | Adjustable quantity via spinner control | High |

- | FR-SALE-04 | Remove individual items or clear entire cart | High |
- | FR-SALE-05 | Payment summary panel: subtotal, tax (IVA), total | High |
- | FR-SALE-06 | Automatic tax calculation per item (based on product tax rate) | High |
- | FR-SALE-07 | Payment method selection: Cash, Credit Card, Debit Card, Transfer | High |
- | FR-SALE-08 | Complete sale: insert sale record, deduct stock, log stock movements | High |
- | FR-SALE-09 | Reject sale if insufficient stock for any item | High |
- | FR-SALE-10 | Recent sales history table (collapsible) | Medium |
- | FR-SALE-11 | Print receipt to ESC/POS thermal printer | Low |

3.6 Purchase Orders (FR-PO)

- | ID | Requirement | Priority |
- |-----|-----|
- | FR-PO-01 | Create purchase orders with: supplier, notes, and line items | High |
- | FR-PO-02 | Each line item has: product, quantity, unit cost, line total | High |
- | FR-PO-03 | Display orders table: ID, date, supplier, total, status, notes | High |
- | FR-PO-04 | Order statuses: PENDING, RECEIVED, CANCELLED | High |
- | FR-PO-05 | Receive order: update status to RECEIVED and add quantities to product stock | High |
- | FR-PO-06 | Cancel order: update status to CANCELLED (no stock change) | Medium |
- | FR-PO-07 | View order items in detail | Medium |

3.7 Reports (FR-RPT)

- | ID | Requirement | Priority |
- |-----|-----|
- | FR-RPT-01 | Report types: Sales Summary, Top Products, Stock Movements, Inventory Value, Profit Margins, Low Stock, Expiring Soon | High |
- | FR-RPT-02 | Date range filter (from/to) | High |
- | FR-RPT-03 | Summary cards: total records, total value, average | High |
- | FR-RPT-04 | Dynamic data table based on report type | High |
- | FR-RPT-05 | Export report to PDF | Medium |
- | FR-RPT-06 | Export report to Excel (.xlsx) | Medium |

3.8 Category Management (FR-CAT)

- | ID | Requirement | Priority |
- |-----|-----|
- | FR-CAT-01 | CRUD operations for categories (name, description) | High |
- | FR-CAT-02 | Display categories in table with edit form | High |

3.9 Supplier Management (FR-SUP)

- | ID | Requirement | Priority |
- |-----|-----|
- | FR-SUP-01 | CRUD operations for suppliers (name, contact, phone, email) | High |
- | FR-SUP-02 | Display suppliers in table with edit form | High |

3.10 User Management (FR-USER)

- | ID | Requirement | Priority |
- |-----|-----|
- | FR-USER-01 | CRUD operations for users (username, display name, role, password, active status) | High |
- | FR-USER-02 | Search users by name | Medium |
- | FR-USER-03 | Prevent deleting own account | High |
- | FR-USER-04 | Password hashing with SHA-256 | High |

3.11 Internationalization (FR-I18N)

- ID

Requirement

Priority
-
- FR-I18N-01

Full Spanish and English interface

High
- FR-I18N-02

Language toggle in sidebar

High
- FR-I18N-03

All view labels, buttons, columns, and messages translated via ResourceBundle

High
- FR-I18N-04

Language preference persisted (Java Preferences API)

Medium

3.12 Backup & Restore (FR-BKP)

- ID

Requirement

Priority
-
- FR-BKP-01

Export SQLite database file as backup

Medium
- FR-BKP-02

Restore database from backup file (replaces current data)

Medium
- FR-BKP-03

Confirmation dialog before restore

High

4. Non-Functional Requirements

4.1 Performance

- ID

Requirement
-
- NFR-01

Application startup in under 3 seconds
- NFR-02

View transitions in under 500ms
- NFR-03

Search results returned in under 200ms for up to 10,000 products

4.2 Security

- ID

Requirement
-
- NFR-04

Passwords stored as SHA-256 hashes (never plain text)
- NFR-05

No external network connections (100% offline)
- NFR-06

Database file stored locally with OS-level file permissions

4.3 Usability

- ID

Requirement
-
- NFR-07

Dark-themed modern UI consistent across all views
- NFR-08

Sidebar navigation with icons and text labels
- NFR-09

Responsive layout adapting to window resize
- NFR-10

Bilingual interface (ES/EN) with instant switching

4.4 Reliability

- ID

Requirement
-
- NFR-11

Database transactions with rollback on failure (sales, purchase orders)
- NFR-12

Connection pooling for database access
- NFR-13

Graceful error handling with user-visible messages

4.5 Portability

- ID

Requirement

|---|-----|

| NFR-14 | Windows 10/11 support via native installer (.exe) |

| NFR-15 | No external dependencies required at runtime (JRE bundled) |

5. Technology Stack

| Component | Technology | Version |

|-----|-----|-----|

| Language | Java | 17+ (JDK 24) |

| UI Framework | JavaFX | 17.0.2 |

| Database | SQLite | 3.x (via sqlite-jdbc) |

| Build Tool | Maven | 3.x |

| PDF Generation | iTextPDF | 5.5.13.4 |

| Excel Export | Apache POI | 5.2.5 |

| Installer | jpackage + WiX | JDK 14+ |

| Thermal Printing | javax.print (ESC/POS) | Java built-in |

6. Database Schema

Tables

- **products** — id, name, description, category_id, supplier_id, price_buy, price_sell, stock_current, stock_min, barcode, tax_rate
- **categories** — id, name, description
- **suppliers** — id, name, contact_person, phone, email
- **users** — id, username, password_hash, display_name, role, active
- **sales** — id, sale_date, user_id, subtotal, tax_amount, total, payment_method, status
- **sale_items** — id, sale_id, product_id, product_name, quantity, unit_price, tax_rate, line_total
- **purchase_orders** — id, order_date, supplier_id, supplier_name, total, status, notes
- **purchase_order_items** — id, order_id, product_id, product_name, quantity, unit_cost, line_total
- **batches** — id, product_id, lot_number, expiry_date, quantity, cost_per_unit, received_date, status
- **stock_movements** — id, product_id, type (IN/OUT), quantity, reference, user_id, created_at
- **price_history** — id, product_id, old_price, new_price, changed_by, changed_at
- **audit_log** — id, action, entity, entity_id, user_id, details, created_at

7. User Interface

7.1 Navigation

Sidebar with icon buttons for: Dashboard, Sales, Products, Purchase Orders, Categories, Suppliers, Reports, Users. Bottom section: language toggle, backup/restore, logout.

7.2 Views

1. **Login** — Username/password form with StockMaster branding
2. **Dashboard** — 4 KPI cards + category bar chart
3. **Sales (POS)** — product search, cart, payment summary, recent sales
4. **Products** — table + edit form + batch panel
5. **Purchase Orders** — orders table + inline order form with items
6. **Categories** — table + edit form
7. **Suppliers** — table + edit form

8. **Reports** — type selector, date filters, summary cards, data table, export buttons

9. **Users** — table + edit form